

Generative AI In Digital Marketing and Content Creation

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Abstract

This project explores the integration of generative artificial intelligence (AI) in digital marketing and content creation, focusing on applications for small businesses. The primary aim is to harness the capabilities of text-to-image generation technologies to streamline and enhance marketing and advertising efforts such as advertisements, social media content, and promotional graphics, at a much reduced costs.

A user-friendly platform, that allows users to input textual descriptions of desired visuals, which the AI then converts into high-quality images using state-of-the-art machine learning model like Stable-Diffusion. The project explores an idea of building a product and developing a possible business model out of such a product.

1. Problem Statement

Small businesses often struggle to create high-quality, engaging marketing and advertising content due to limited resources, lack of design expertise, and budget constraints. Traditional content creation processes are time-consuming, expensive, and require specialised skills that small business owners may not possess or afford. This results in suboptimal marketing efforts, reduced audience engagement, and ultimately, hindered business growth. At this age of internet missing out on the opportunity to do promotions on social media platforms would be disadvantageous for the businesses.

I, thus try to tackle this problem with Generative AI, specifically a text-to-image generation application, which offers a promising solution to these challenges by enabling the automated creation of visually appealing and customised marketing materials at a higher degree of control, customisability, much reduced cost and efforts.

2. Market/ Customer Needs Assessment

2.1 Market Analysis

The global digital marketing industry is projected to grow at a compound annual growth rate (CAGR) of 17.6% from 2021 to 2026. Small businesses constitute a significant portion of this market, with millions of SMEs globally seeking effective digital marketing solutions, which can help them compete with larger enterprises.

There is an increasing demand for high-quality digital content, with visual content being a critical component, which can get up to 94% more views than content without them. The rise of social media platforms and visual-based marketing channels like Instagram and Pinterest highlights the need for visually appealing content.

2.2 Customer Segmentation

Typically, small business owners have limited time, resources and have tight marketing budgets and cannot afford expensive design software or hiring professional designers. They need affordable, easy-to-use tools to create marketing content without requiring extensive design skills. There is a need for intuitive tools that can produce high-quality visuals with minimal input.

Businesses such as local restaurants, boutiques, and service providers require regular updates to their marketing content to engage with their community and attract new customers. They need tools that allow for quick turnaround times and flexibility in content creation.

Maintaining a consistent brand image across all marketing channels is challenging without professional design help. Small businesses need tools that help them easily incorporate their brand elements (logos, colours, fonts) into their content. Automated and quick content generation is essential to allow them to focus on other critical aspects of their business.

3. Target Specification

3.1 Core Functionality and Design

Text-to-Image Generation: Converts user input text descriptions into high-quality images. Supports various content types (advertisements, social media posts, promotional graphics).

Customisation: Allows users to add logos, brand colours, fonts, and other brand elements. Provides editing tools for further image refinement.

Intuitive Design: Simple, user-friendly interface suitable for users with no design experience.

3.2 Performance Requirements

Speed and Efficiency: Text-to-image generation should be completed within a short period of time to ensure a smooth user experience. 99.9% uptime to ensure reliability for users relying on the platform for regular content creation.

Scalability: Must handle multiple users simultaneously without performance degradation. Scalable infrastructure to support growing user base and increased content generation demands.

Quality: Generated images must be of high quality suitable for various digital platforms. Consistency in visual quality across different types of content.

4. External Search

The product requires usage of deep learning models, which are really hard to write from scratch in order to suit our needs. Tweaking these models for our use is less daunting than coding it up from scratch. A pre-trained model can be used with the help of transfer learning methods. A lot of such pre-trained model are there which are open source and available in <https://huggingface.co/models> . One such model based on Stable Diffusion technique by StabilityAI can be used.

4.1. Benchmarking

In the market of text-to-image generation there are several already available models like DALL-E, DALL-E 2.0 from OpenAI. There is also Midjourney based on Stable Diffusion which are not free. My idea is to build a text-to-image generator product which, even though can be repurposed for any task, is well suited for small businesses and for integration to any software system through API.

For the model itself, I have used “stable-diffusion-v1-4” from CompVis which is available in Huggingface. This particular model showed better stability in image generation compared to others in my system.

5. Constraints and Regulations

Several constraints must be kept in mind-

- The image generation depends on server performance, thus a high performance server is required.
- The model automatically filters out explicit content which is an advantage.
- Data protection and privacy regulations(Customers).
- Govt Regulations for small businesses.

6. Business Model

A business and monetisation model can be developed around such a product geared towards small businesses and even for individual users.

A subscription based model would be well suited for such a product. A tiered subscription plan e.g Basic, Pro, Enterprise to cater to different business sizes and needs and even individual creators, users. Competitive pricing to ensure affordability for small businesses. Offer a free tier with basic features and limited image generation to attract users. Encourage users to upgrade to paid plans for more features and higher usage limits.

A Basic plan with limited features and a cap on the number of images generated per month. Ideal for solo entrepreneurs and very small businesses.

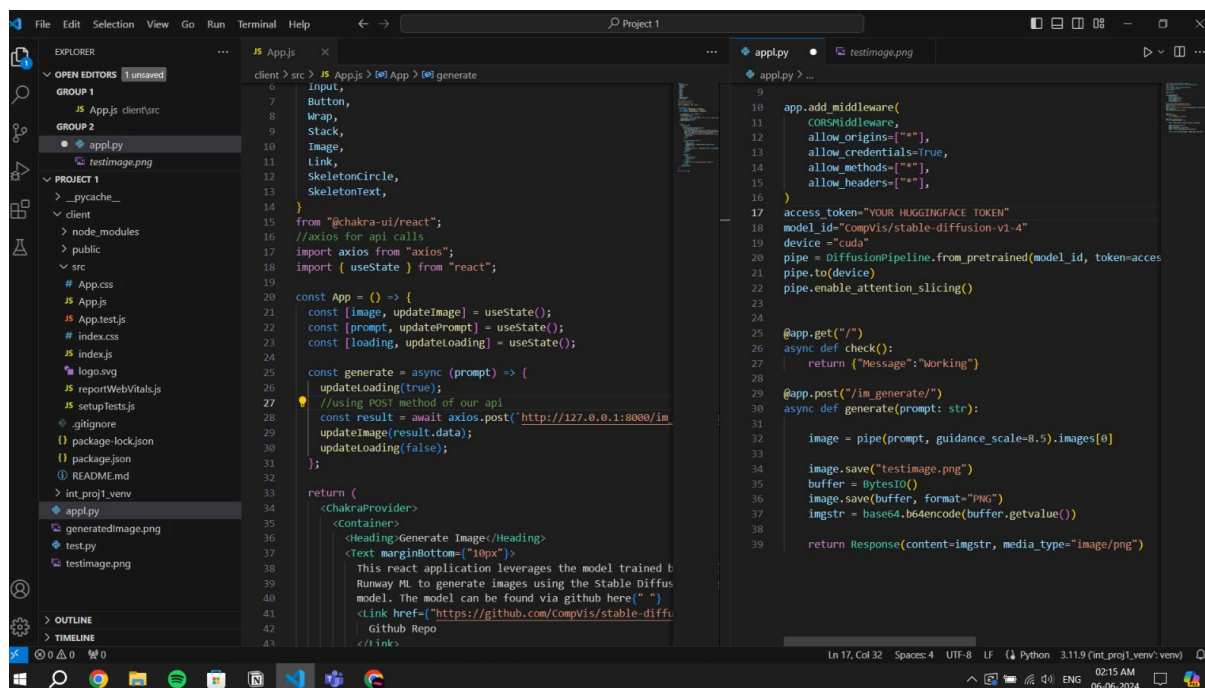
A Pro plan with unlimited image generation, advanced customisation options, and priority customer support. Suitable for growing small businesses and freelancers.

An Enterprise plan with custom solutions, dedicated account manager, team collaboration features, and API access. Designed for marketing agencies and larger small businesses.

In-platform advertising can be used Display non-intrusive ads within the platform, such as relevant marketing tools, services and also sponsorships.

7. Concept Generation and Development

In order to build such a product a usual and simple front-end interface with a high performance back-end with availability of API integration can be a viable option. Huggingface is a platform where, there are a number of pre-trained models available.

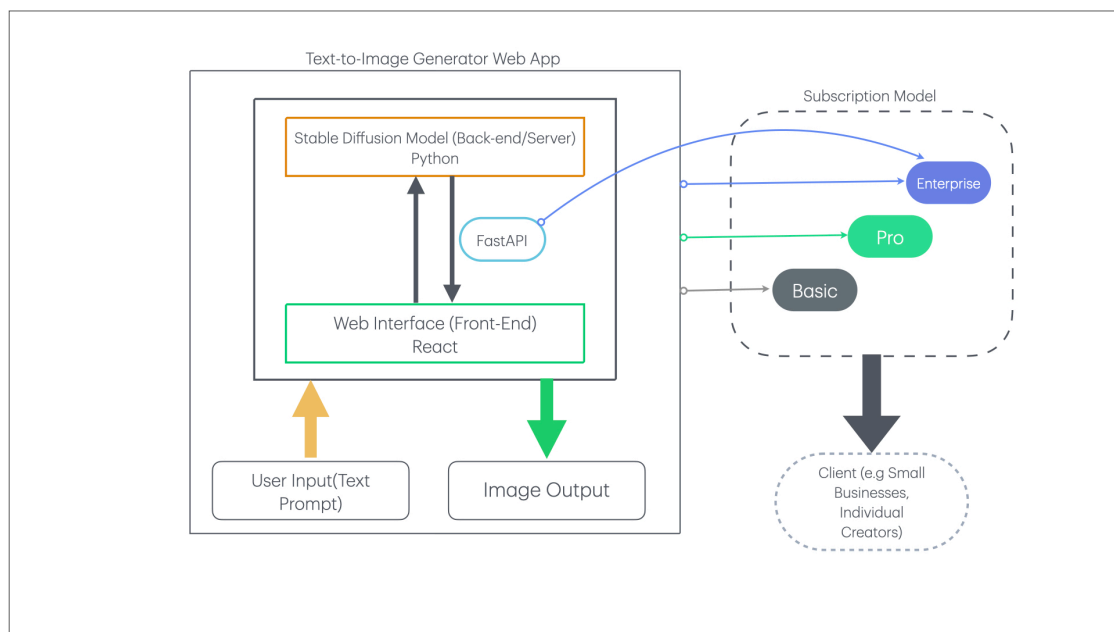


A portion of the Frontend and Python backend code

The Stable Diffusion model runs in a server which acts as a backend and a front-end or a software component can provide text prompt and obtain the generated image. The frontend is built using React framework in JavaScript and Backend using python and FastAPI according to the need.

8. Final Product Prototype

The product has the following components and functions to provide satisfactory results.



Schematic Diagram of the Final Product Prototype

Front-End: The front-end interface, which the client will interact with, is build using React Framework and JavaScript. It includes a basic Header and Input text field and a Generate button which triggers the Back-end through API. The received JSON response has the encoded image, which is decoded and displayed on the client side interface.

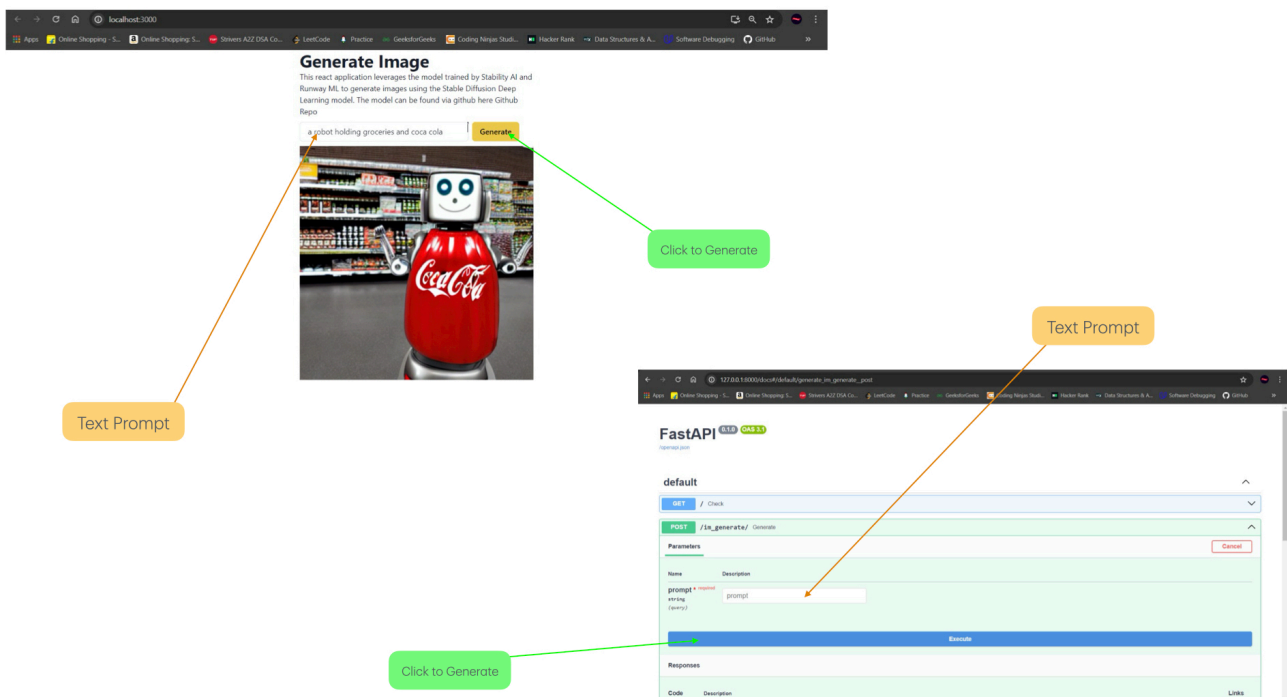
Back-End: The back-end is build in python language and FastAPI is used for building and generating the api interface. The Generate button in front-end triggers the server side python code which generates an image using the Stable Diffusion model and returns an image encoded in JSON. The Stable Diffusion uses PyTorch.



9. How Does it Work?

The client side will have an interface to input text and a generate button to start generation. To access for client side-

- The client needs to input the prompt.
- Click the generate button
- Generated Picture appears below

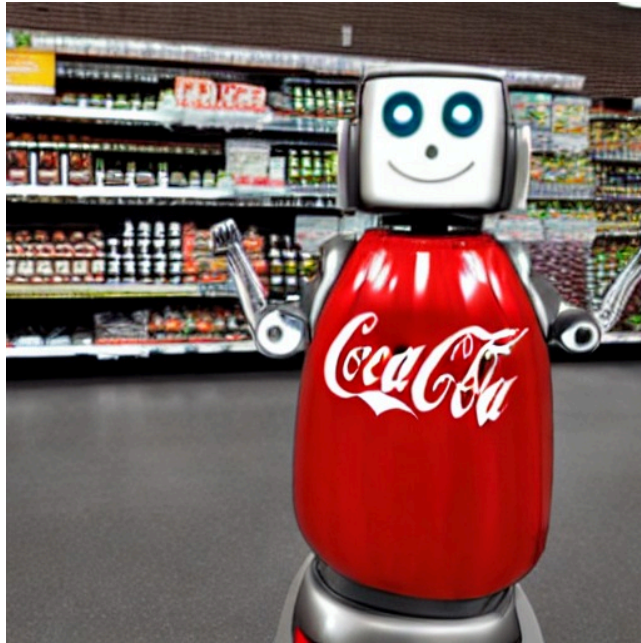


To access using api-

- At the given api link, user will get a FastAPI interface as above.
- User needs to click "Try Out" at the POST method and give the prompt.
- Then click the execute and the image will be generated at server side.

10. Implementation

Using the web app developed, I have been able to generate various images giving text prompt input. Some sample images are given below-



Prompt: a robot holding groceries and coca cola



Prompt: a cycle

The code implementation is uploaded in my GitHub.

The GitHub link- https://github.com/iamabhijit01/feynnlabs_project01_text2image

11. Conclusion

The AI-Marketer project revolutionises digital marketing for small businesses by using generative AI for text-to-image creation. This user-friendly platform addresses the challenges of limited resources, design expertise, and time, enabling small businesses to produce high-quality, customised marketing content easily and affordably.

With a robust business model featuring subscription plans, pay-per-use options, and premium features, AI-Marketer ensures sustainable revenue and growth. By democratising access to professional-grade marketing tools, AI-Marketer empowers small businesses to enhance their digital marketing efforts, engage audiences effectively, and drive business success.

12. References and Resources

1. Rombach, Robin, et al. "High-resolution image synthesis with latent diffusion models." Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2022. <https://arxiv.org/abs/2112.10752>
2. R. Rojas, *Neural networks: a systematic introduction*. Berlin, Heidelberg: Springer-Verlag, 1996.
2. <https://huggingface.co/CompVis/stable-diffusion-v1-4>
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